Supporting Information

When Does Generator Metadata Help? A Controlled Study of Structure-Aware Row Scaling for Generated Mixed-Integer Programs

Mathias Rodríguez Castro

This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1); (iv) the per-variant original-scale reliability breakdown with PAR10 under the three readings of the main text (Table S2); (v) the out-of-sample sensitivity of the collective coupling reliability floor (Table S3); (vi) the evaluation of the optional local reliability floor on the failing Simple subset (Table S4); (vii) the common-pool exclusions and full-pool sensitivity (Table S5); (viii) paired-test attrition, the corrected Wilcoxon procedure, and clustered-bootstrap accounting (Table S6); (ix) the available solver-internal-scaling sweep and the verified CPLEX `Read::Scale= −1` audit (Table S7); and (x) the archived runtime comparison used for the coupling-stage activation audit (Table S8); (xi) the direct SA–Flat strict/mixed reliability audit (Table S9); and (xii) the multiplicity, operational-family, bootstrap, preprocessing-plus-solver, deterministic-effort, and own-incumbent sensitivities for that direct comparison (Table S10); (xiii) its per-contrast attrition and cached-metadata endpoint on both builds (Table S11); and (xiv) the measurement environment, build comparison and cross-environment transfer-factor sensitivity (Table S12); and (xv) the full traceability of the kernel-matched metadata grid— the block-heterogeneous family, its per-block scale distribution, the seeds, the per-variant κ_2^+ distribution on the headline cell, and the reproduction command (Section S3, Table S13).

Section S1. Proofs of the spectral propositions

These are the full proofs of the two propositions stated in Section 3 of the main text, moved here to keep the motivation and the method adjacent.

Proof of Proposition 1 (Ideal scalar scaling of block-diagonal matrices).

Proof. For fixed positive scalars α_i, the scaled matrix is block diagonal:

$$\mathrm{diag}(\alpha_1 I, \ldots, \alpha_B I)D = \mathrm{diag}(\alpha_1 A_1, \ldots, \alpha_B A_B).$$

The singular values of a block-diagonal matrix are the union of the singular values of its blocks. Therefore, for every i,

$$\kappa_2(\mathrm{diag}(\alpha_1 A_1, \ldots, \alpha_B A_B)) \geq \kappa_2(\alpha_i A_i) = \kappa_2(A_i).$$

Taking the maximum over i gives the lower bound

$$\kappa_2(\mathrm{diag}(\alpha_1 A_1, \ldots, \alpha_B A_B)) \geq \max_i \kappa_2(A_i)$$

for every admissible block scaling.

Now choose $\alpha_i = (M_i m_i)^{-1/2}$. Then the largest and smallest singular values of $\alpha_i A_i$ are

$$\sigma_{\max}(\alpha_i A_i) = \sqrt{\frac{M_i}{m_i}} = \sqrt{\kappa_2(A_i)},$$
$$\sigma_{\min}(\alpha_i A_i) = \sqrt{\frac{m_i}{M_i}} = \frac{1}{\sqrt{\kappa_2(A_i)}}.$$

Because the singular values of the complete block-diagonal matrix are obtained by pooling the singular values of its blocks,

$$\sigma_{\max}(\mathrm{diag}(\alpha_1 A_1, \ldots, \alpha_B A_B)) = \sqrt{\max_i \kappa_2(A_i)},$$

and

$$\sigma_{\min}(\mathrm{diag}(\alpha_1 A_1, \ldots, \alpha_B A_B)) = \frac{1}{\sqrt{\max_i \kappa_2(A_i)}}.$$

The resulting condition number is $\max_i \kappa_2(A_i)$, matching the lower bound. □

Proof of Proposition 2 (Conditioning of a triangular coupling factor).
Proof. Let $R = U\Sigma V^\top$ be a singular value decomposition (SVD) of R, with U and V orthogonal. Orthogonal multiplication does not change singular values, and hence T_R has the same singular values as

$$\begin{bmatrix} V^\top & 0 \\ 0 & U^\top \end{bmatrix} \begin{bmatrix} I_n & 0 \\ R & I_m \end{bmatrix} \begin{bmatrix} V & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ \Sigma & I_m \end{bmatrix}.$$

For each singular value σ of R, the nontrivial part reduces to the two-dimensional block

$$B_\sigma = \begin{bmatrix} 1 & 0 \\ \sigma & 1 \end{bmatrix}.$$

The squared singular values of B_σ are the eigenvalues of

$$B_\sigma^\top B_\sigma = \begin{bmatrix} 1+\sigma^2 & \sigma \\ \sigma & 1 \end{bmatrix},$$

namely

$$\lambda_\pm(\sigma) = \frac{\sigma^2 + 2 \pm \sigma\sqrt{\sigma^2+4}}{2}.$$

Their product is one, and $\lambda_+(\sigma)$ is increasing for $\sigma \geq 0$. Additional directions associated with zero singular values of R contribute singular values equal to one. Therefore the largest squared singular value of T_R is $\lambda_+(\rho)$ and the smallest squared singular value is $1/\lambda_+(\rho)$. Consequently,

$$\begin{aligned} \kappa_2(T_R) &= \lambda_+(\rho) \\ &= \frac{\rho^2 + 2 + \rho\sqrt{\rho^2+4}}{2} \\ &= \left(\frac{\sqrt{\rho^2+4}+\rho}{2}\right)^2. \end{aligned}$$

□

Section S2. Supplementary computational results

Here, PAR10 (including its ITT/strict/mixed superscripts) denotes the common-pool metric defined in the main text. The distinct symbol $\mathrm{PAR10}_{\mathrm{avail}}$ is reserved for archived descriptive revision tables: its denominator is the variant's available solver-output records, a timeout receives 36,000 s, and a pure pipeline error with no solver output is excluded. It must not be compared numerically with the fixed-pool metric.

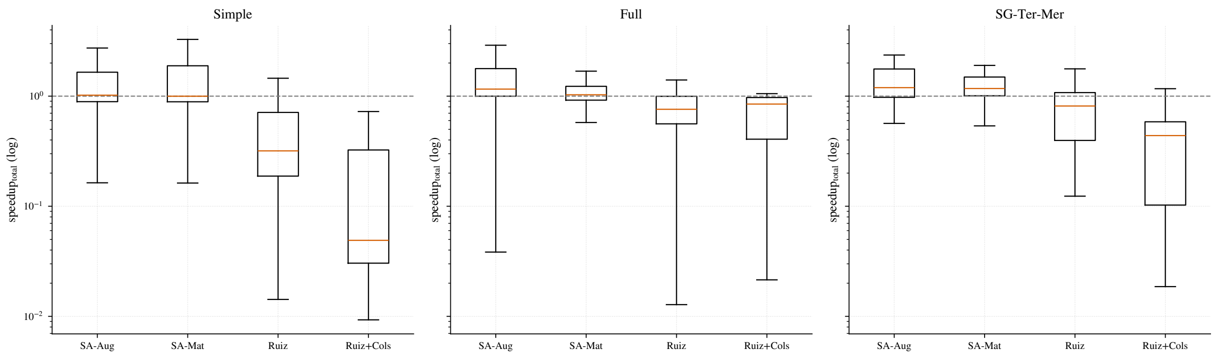
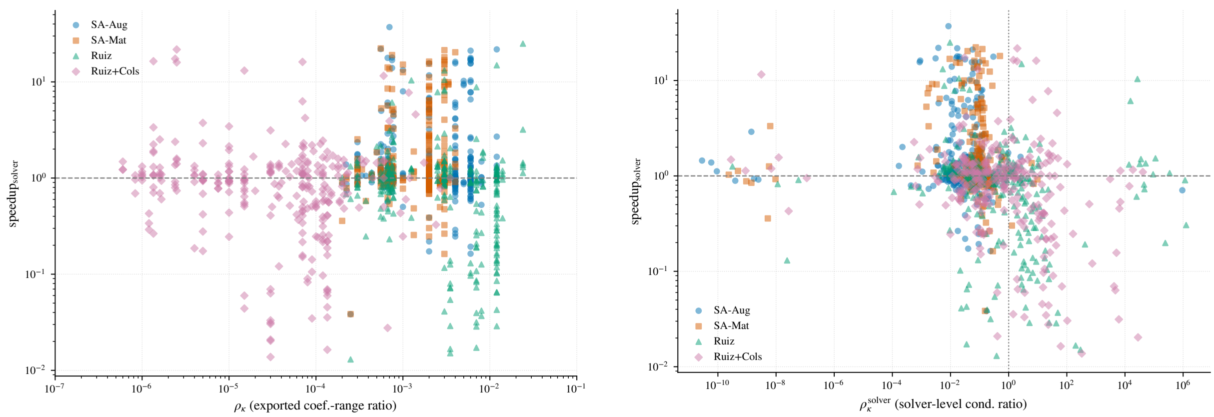


Figure S1: At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.



(a) Static coefficient-range ratio.

(b) Solver-level conditioning ratio.

Figure S2: At 0.1% MIPGap, the pooled scatter shows a stronger association with speedup for the own-incumbent path-specific fixed-LP diagnostic than for the exported-range proxy. This describes solver paths; it does not make the former a purer measure of representation or an instance-level predictor for the structure-aware variants.

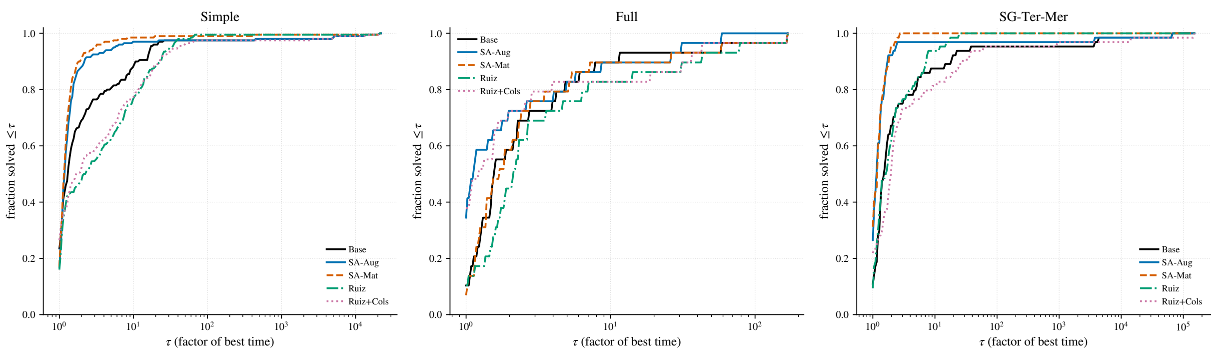
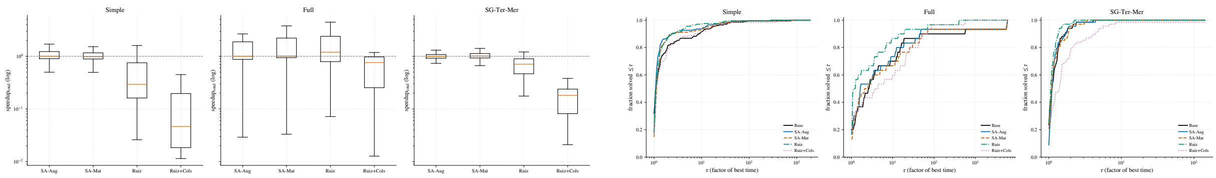


Figure S3: At 0.1% MIPGap, fixed-pool Dolan–Moré profiles use the common denominators and individual 36,000-s failure cost defined in the main text.



(a) Total-time speedup distribution.

(b) Fixed-pool performance profile.

Figure S4: At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance. The profile uses a common pool and assigns 36,000 s to an individual timeout, abort, missing run, or invalid time, as defined in the main text.

Table S1: Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm ($s = 1$ s) and descriptive $\mathrm{PAR10}_{\mathrm{avail}}$, in seconds; default settings in bold.

Parameter	Value	sgm (s)	$\mathrm{PAR10}_{\mathrm{avail}}$ (s)
	1.5	2.09	2.46
Band u	**2**	2.07	2.45
	4	2.12	2.49
	$[10^{-6}, 10^{6}]$	2.04	2.40
Window $[\varepsilon_{\min}, \varepsilon_{\max}]$	$[\mathbf{10^{-9}}, \mathbf{10^{9}}]$	2.02	2.39
	$[10^{-12}, 10^{12}]$	2.01	2.36

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings (sgm $\in [2.01, 2.12]$) is far below the base-to-scaled gap, and the median paired speedup is ≈ 1.0 throughout. The number of Ruiz iterations K is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at $K = 4$; varying it leaves SA-Aug unchanged, as expected.

Table S2: Per-variant audit and PAR10 under the three operational readings. All campaigns use the manifest-declared $T_{\max} = 3600$ s. A timeout or pipeline abort receives $10T_{\max} = 36000$ s in every reading; a returned output that fails the relevant verifier receives that penalty only in the strict or mixed reading. The count columns use the raw manifest pool: n_{att} is every attempted instance and n_{out} reproduces the pooled diagnostic audit's status-OK denominator, including time-limited runs and three records that lack verification fields. TO is a subset of n_{out}, and Abort is $n_{\text{att}} - n_{\text{out}}$. Pass counts are among those status-OK records; missing verification is non-passing. Strict pass means $v_{\text{row}} \le 10^{-4}$ and $v_{\text{int}} \le 10^{-5}$; mixed pass rescues only row failures with $v_r/(1 + |b_r|) \le 10^{-6}$. In contrast, the four PAR10 columns use the reproducible scenario-specific common pool after the uniformly-unavailable exclusion in Table S5: Base ITT and, for the listed variant, ITT/strict/mixed. Values are seconds, rounded to the nearest second.

Solver	Gap	Class	Variant	n_{att}	n_{out}	TO	Abort	$n^{\text{str}}_{\text{pass}}$	$n^{\text{mix}}_{\text{pass}}$	Base ITT	ITT	Strict	Mixed
Gurobi	1%	Simple	SA-Aug	204	197	0	7	185	185	373	364	2535	2535
Gurobi	1%	Simple	SA-Mat	204	198	0	6	179	180	373	183	3620	3439
Gurobi	1%	Simple	Ruiz	204	198	0	6	143	146	373	207	10153	9611
Gurobi	1%	Simple	Ruiz+Cols	204	199	0	5	180	189	373	28	3462	1835
Gurobi	1%	SG-Ter-Mer	SA-Aug	68	67	1	1	66	67	1094	1083	1613	1083
Gurobi	1%	SG-Ter-Mer	SA-Mat	68	67	0	1	65	66	1094	1079	1608	1079
Gurobi	1%	SG-Ter-Mer	Ruiz	68	68	1	0	67	68	1094	567	1095	567
Gurobi	1%	SG-Ter-Mer	Ruiz+Cols	68	68	1	0	67	68	1094	585	1113	585
Gurobi	1%	Full	SA-Aug	34	34	3	0	27	27	3385	3308	9640	9640
Gurobi	1%	Full	SA-Mat	34	34	2	0	31	31	3385	3473	5586	5586
Gurobi	1%	Full	Ruiz	34	33	4	1	31	31	3385	5491	7577	7577
Gurobi	1%	Full	Ruiz+Cols	34	33	4	1	31	32	3385	5617	6666	5617
Gurobi	0.1%	Simple	SA-Aug	204	195	0	9	174	178	928	928	4705	3987
Gurobi	0.1%	Simple	SA-Mat	204	198	0	6	168	172	928	381	5776	5057
Gurobi	0.1%	Simple	Ruiz	204	199	0	5	155	159	928	221	8138	7419
Gurobi	0.1%	Simple	Ruiz+Cols	204	195	0	9	175	185	928	948	4544	2746
Gurobi	0.1%	SG-Ter-Mer	SA-Aug	68	63	1	5	62	63	2246	1683	2237	1683
Gurobi	0.1%	SG-Ter-Mer	SA-Mat	68	65	0	3	63	64	2246	576	1130	576
Gurobi	0.1%	SG-Ter-Mer	Ruiz	68	65	1	3	65	65	2246	587	587	587
Gurobi	0.1%	SG-Ter-Mer	Ruiz+Cols	68	62	1	6	61	62	2246	2262	2814	2262
Gurobi	0.1%	Full	SA-Aug	34	30	3	4	27	27	5992	5897	8140	8140
Gurobi	0.1%	Full	SA-Mat	34	31	4	3	28	30	5992	6026	9293	7148
Gurobi	0.1%	Full	Ruiz	34	30	4	4	27	27	5992	7066	9277	9277
Gurobi	0.1%	Full	Ruiz+Cols	34	31	5	3	28	29	5992	7073	9312	8191
CPLEX	1%	Simple	SA-Aug	204	204	0	0	37	37	6	3	29471	29471
CPLEX	1%	Simple	SA-Mat	204	204	0	0	43	43	6	3	28412	28412
CPLEX	1%	Simple	Ruiz	204	204	0	0	44	44	6	4	28237	28237
CPLEX	1%	Simple	Ruiz+Cols	204	204	0	0	137	138	6	3	11825	11649
CPLEX	1%	SG-Ter-Mer	SA-Aug	68	68	0	0	68	68	3	10	10	10
CPLEX	1%	SG-Ter-Mer	SA-Mat	68	68	0	0	68	68	3	6	6	6
CPLEX	1%	SG-Ter-Mer	Ruiz	68	68	0	0	68	68	3	10	10	10
CPLEX	1%	SG-Ter-Mer	Ruiz+Cols	68	68	0	0	68	68	3	15	15	15

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Table S2 (continued)

Solver	Gap	Class	Variant	n_{att}	n_{out}	TO	Abort	$n_{\text{pass}}^{\text{str}}$	$n_{\text{pass}}^{\text{mix}}$	Base ITT	ITT	Strict	Mixed
CPLEX	1%	Full	SA-Aug	34	34	5	0	25	25	6515	5546	15014	15014
CPLEX	1%	Full	SA-Mat	34	34	6	0	25	26	6515	6498	14932	13875
CPLEX	1%	Full	Ruiz	34	34	3	0	28	28	6515	3506	8744	8744
CPLEX	1%	Full	Ruiz+Cols	34	34	4	0	32	33	6515	4681	6793	5737
CPLEX	0.1%	Simple	SA-Aug	204	204	1	0	40	40	214	188	29119	29119
CPLEX	0.1%	Simple	SA-Mat	204	204	1	0	38	38	214	203	29311	29311
CPLEX	0.1%	Simple	Ruiz	204	204	1	0	44	44	214	192	28413	28413
CPLEX	0.1%	Simple	Ruiz+Cols	204	204	3	0	144	146	214	547	10955	10779
CPLEX	0.1%	SG-Ter-Mer	SA-Aug	68	68	0	0	68	68	19	21	21	21
CPLEX	0.1%	SG-Ter-Mer	SA-Mat	68	68	0	0	68	68	19	16	16	16
CPLEX	0.1%	SG-Ter-Mer	Ruiz	68	68	0	0	68	68	19	8	8	8
CPLEX	0.1%	SG-Ter-Mer	Ruiz+Cols	68	68	1	0	68	68	19	539	539	539
CPLEX	0.1%	Full	SA-Aug	34	34	8	0	24	24	9951	8930	18283	18283
CPLEX	0.1%	Full	SA-Mat	34	33	10	1	25	26	9951	11916	20280	19226
CPLEX	0.1%	Full	Ruiz	34	34	5	0	32	32	9951	5857	7917	7917
CPLEX	0.1%	Full	Ruiz+Cols	34	33	10	1	31	32	9951	11950	14063	13007

Objective-difference audit. MIPGap bounds each incumbent against its own dual bound and is not a cross-run agreement test. Of 4,806 variant runs with returned solver output, 72 differ from the corresponding Base incumbent by more than twice the nominal gap; these cases are concentrated in a small set of instance identifiers. The largest pooled relative deviations occur in Full/Ruiz cells ($|\Delta z_{\mathrm{rel}}| \approx 1.8 \times 10^2$ under Gurobi and 5.3×10^1 under CPLEX). Base is not presumed correct: one Simple Base incumbent is 73% worse than the best verified incumbent, and one CPLEX Full Base incumbent is roughly six times worse. Accordingly, correctness is assessed against the best original-scale-verified incumbent on each instance, with Base subjected to the same feasibility and integrality checks as every scaled variant.

Table S3: Out-of-sample sensitivity of the reliability floor for the collective coupling factor (SA-Aug), on twenty seeds (21–40) never used while designing the floor. Cells: coupling, coupling-uniform, and mixed patterns at severity $S = 6$ (60 instances, Gurobi, MIPGap 10^{-3}, 600 s limit). Median post-scaling global coefficient-and-right-hand-side magnitude-range proxy per pattern; number of runs failing original-scale verification; worst original-scale row violation; instances whose objective disagrees with the base beyond the MIPGap. The failure mode replicates out-of-sample with the floor disabled, and every floor value across three orders of magnitude eliminates it.

Floor	coupling	coupling-unif.	mixed	n_{fail}	$v_{\max}$	over gap
(pre-scaling)	4.4×10^8	3.3×10^6	1.5×10^{14}			
disabled	3.8×10^7	5.4×10^3	2.9×10^7	4	4.2×10^{-1}	4
10^{-2}	9.1×10^5	5.4×10^3	2.9×10^6	0	1.2×10^{-7}	0
10^{-3} (default)	3.5×10^5	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-4}	1.0×10^6	5.4×10^3	2.3×10^6	0	1.2×10^{-7}	0
10^{-5}	2.1×10^6	5.4×10^3	3.0×10^6	0	1.2×10^{-7}	0

Coupling-trigger sensitivity. On the original synthetic grid, sweeping $\widehat{\rho}_{\mathrm{trigger}} \in \{2, 5, 10, 20, 50\}$ leaves every displayed coupling and coupling-uniform cell median unchanged. Approximately one instance in twenty lies in the $[2, 50]$ band, so some individual activation decisions change, but the proxy is effectively bimodal (above 10^2 on oversized instances and below one on undersized instances). The result therefore supports the stated conclusion about the one-sided design, not a privileged choice of the cutoff 10.

Table S4: Optional local reliability floor (SA-Aug-LF), evaluated on the twelve Simple instances (Gurobi, 1% MIPGap) whose SA-Aug runs fail original-scale verification. The floor applies the local factor only if $\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}}$, so $d_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$ bounds the conditional original-scale residual by $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$; rows below the floor stay at base scale. n_{fail} is the number of the twelve runs failing the strict absolute criterion; σ_{gm} is the shifted geometric mean solver time ($s = 1$ s); $\widehat{\kappa}_{\text{range}}^{\text{post}}$ is the median exported coefficient-and-right-hand-side magnitude-range proxy; $v_{\max}$ is the worst original-scale row violation. SA-Mat is shown for reference.

Variant	n_{fail}	σ_{gm} (s)	$\widehat{\kappa}_{\text{range}}^{\text{post}}$	$v_{\max}$
Base	0	4.90	–	0
SA-Aug (no floor, default)	12	2.47	6.6×10^{9}	3.7×10^{-3}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-1}$	0	2.64	5.5×10^{11}	5.3×10^{-5}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$	0	2.89	5.5×10^{11}	7.4×10^{-5}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-3}$	6	5.52	2.0×10^{14}	4.0×10^{-2}
SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-4}$	6	2.19	2.0×10^{14}	5.3×10^{-4}
SA-Mat	5	2.57	4.5×10^{12}	1.9×10^{-3}

Notes. The floor is monotone: n_{fail} falls from 12 (no floor) to 6 (shallow floors $10^{-3}/10^{-4}$, which protect only the most extreme rows) to 0 ($10^{-2}/10^{-1}$), consistent with the residual bound $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$. At the effective floor 10^{-2} the variant eliminates every failure while retaining a solver-time advantage over the base (σ_{gm} 2.89 vs 4.90 s), even though it *raises* the exported magnitude-range proxy (the protected rows are exactly the over-compressed large-right-hand-side rows)—a further illustration that this static proxy is a weak predictor of runtime. The threshold is imposed on the representation-dependent factor $\alpha_r^{A,b}$ and is therefore not invariant to a positive re-emission of the same unscaled row. Here it is defined relative to the generator's canonical export in physical units. A representation-invariant alternative could threshold a norm of the resulting scaled row, but it would not inherit the same $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ residual bound. The σ_{gm} values on twelve instances are noisy and are not a full-class estimate. This subset was selected by a post-treatment outcome (the SA-Aug failures), so the table is a demonstration on the failing set; the full-class reliability and runtime effect, and the behavior on Full, remain untested.

Table S5: Uniformly unavailable operational records and sensitivity to retaining them. Panel A applies the documented reproducible data rule: exclude an instance only when all five attempted variants have nonzero pipeline return codes. Panel B restores those cells to the manifest-declared full pool and assigns each a 36000-s penalty. Triplets are ITT/strict/mixed PAR10 in seconds; only scenarios with exclusions are shown because common- and full-pool results otherwise coincide. The within-reading ordering of Base, SA-Aug, and SA-Mat is unchanged (the rounded common-pool tie in Gurobi Simple at 0.1% is broken by less than one second before rounding in the full pool).

Panel A: identifiers

Solver	Gap	Class	N/n	Uniformly unavailable identifiers
Gurobi	1%	Simple	204/199	`caso46f`, `caso47a`–`caso47d`
Gurobi	0.1%	Simple	204/200	`caso02c`–`caso02e`, `caso22a`
Gurobi	0.1%	SG-Ter-Mer	68/65	`instance020`–`instance022`
Gurobi	0.1%	Full	34/32	`instance046`, `instance047`

Panel B: common pool versus penalized full pool

Gap/class	Variant	Pool	ITT	Strict	Mixed
1%, Simple (n = 199, N = 204)	Base	common	373	554	554
		full	1246	1422	1422
	SA-Aug	common	364	2535	2535
		full	1238	3355	3355
	SA-Mat	common	183	3620	3439
		full	1061	4414	4237
	Ruiz	common	207	10153	9611
		full	1085	10786	10258
	Ruiz+Cols	common	28	3462	1835
		full	909	4260	2673
0.1%, Simple (n = 200, N = 204)	Base	common	928	1108	1108
		full	1615	1792	1792
	SA-Aug	common	928	4705	3987
		full	1616	5318	4615
	SA-Mat	common	381	5776	5057
		full	1079	6369	5664
	Ruiz	common	221	8138	7419
		full	923	8684	7980
	Ruiz+Cols	common	948	4544	2746
		full	1635	5161	3398
0.1%, SG-Ter-Mer (n = 65, N = 68)	Base	common	2246	2246	2246
		full	3735	3735	3735
	SA-Aug	common	1683	2237	1683
		full	3197	3727	3197
	SA-Mat	common	576	1130	576
		full	2139	2668	2139
	Ruiz	common	587	587	587
		full	2149	2149	2149
	Ruiz+Cols	common	2262	2814	2262
		full	3750	4279	3750
0.1%, Full (n = 32, N = 34)	Base	common	5992	5992	5992
		full	7757	7757	7757
	SA-Aug	common	5897	8140	8140
		full	7668	9779	9779
	SA-Mat	common	6026	9293	7148
		full	7789	10864	8845
	Ruiz	common	7066	9277	9277
		full	8768	10849	10849
	Ruiz+Cols	common	7073	9312	8191
		full	8775	10882	9827

Note. N is the manifest pool and n the common pool. In the archived logs for every excluded identifier, Base, SA-Aug, Ruiz, and Ruiz+Cols stop at a Gurobi WLS `Overage for too long` licensing error and report no configured solver. The SA-Mat logs instead record optimal completions written to a separate `*-matricial` result tree, while their aggregate campaign records retain nonzero return codes. Thus the evidence supports a combination of license-capacity failure and incomplete artifact consolidation; the exclusion itself is defined only by the observable all-variant return-code rule and does not assert that every underlying solve failed.

Table S6: Pair attrition and corrected Wilcoxon audit. n is the effective matched-pair count. UM denotes the all-variant uniformly-unavailable exclusion; D_{TO}, D_{A}, and D_{mix} are exclusions because both Base and the listed variant are, respectively, timeouts, pipeline aborts, or different kinds of noncompletion. C_1 is a one-sided noncompletion retained at exactly $T_{\max} = 3600\,\mathrm{s}$. There are no missing-time or exact-zero pairs in any row. A one-sided timeout is right-censored at $T_{\max}$; assigning the same cost to a one-sided algorithmic/pipeline abort is an operational-failure convention rather than a censoring claim. The test is two-sided Wilcoxon signed-rank on $t_{\mathrm{variant}} - t_{\mathrm{Base}}$ (SciPy 1.15.3), with exact zeros discarded, tied absolute differences assigned midranks, and no continuity correction. The exact null distribution is used for at most 50 nonzero untied differences; otherwise the tie-corrected normal approximation is used. The displayed structure-aware Simple/SG rows use the approximation and Full rows use the exact distribution; no zero or tied-absolute-difference pair occurs. $r_{\mathrm{rb}} < 0$ favors the listed variant. q is Benjamini–Hochberg adjusted over the twelve tests in each solver-by-gap family (three classes by four variants).

Solver	Gap	Class	Variant	n	UM	D_{TO}	D_{A}	D_{mix}	C_1	W	p	q	r_{rb}
Gurobi	1%	Simple	SA-Aug	197	5	0	2	0	0	7535	5.67×10^{-3}	9.72×10^{-3}	−0.227
Gurobi	1%	Simple	SA-Mat	198	5	0	1	0	1	3011	2.42×10^{-17}	2.90×10^{-16}	−0.694
Gurobi	1%	Simple	Ruiz	198	5	0	1	0	1	7248	1.27×10^{-3}	3.04×10^{-3}	+0.264
Gurobi	1%	Simple	Ruiz+Cols	199	5	0	0	0	2	7414	1.82×10^{-3}	3.65×10^{-3}	+0.255
Gurobi	1%	SG-Ter-Mer	SA-Aug	66	0	1	1	0	0	501	1.13×10^{-4}	4.51×10^{-4}	−0.547
Gurobi	1%	SG-Ter-Mer	SA-Mat	66	0	0	1	1	0	115	2.49×10^{-10}	1.50×10^{-9}	−0.896
Gurobi	1%	SG-Ter-Mer	Ruiz	67	0	1	0	0	1	1129	0.950	0.950	−0.009
Gurobi	1%	SG-Ter-Mer	Ruiz+Cols	67	0	1	0	0	1	598	7.26×10^{-4}	2.18×10^{-3}	+0.475
Gurobi	1%	Full	SA-Aug	32	0	2	0	0	2	175	0.0984	0.148	−0.337
Gurobi	1%	Full	SA-Mat	32	0	2	0	0	2	204	0.270	0.360	−0.227
Gurobi	1%	Full	Ruiz	31	0	2	0	1	2	197	0.327	0.393	+0.206
Gurobi	1%	Full	Ruiz+Cols	32	0	2	0	0	4	243	0.705	0.769	+0.080
Gurobi	0.1%	Simple	SA-Aug	195	4	0	5	0	0	6963	1.02×10^{-3}	3.06×10^{-3}	−0.271
Gurobi	0.1%	Simple	SA-Mat	198	4	0	2	0	3	7069	5.70×10^{-4}	2.28×10^{-3}	−0.282
Gurobi	0.1%	Simple	Ruiz	199	4	0	1	0	4	7429	1.94×10^{-3}	4.66×10^{-3}	+0.253
Gurobi	0.1%	Simple	Ruiz+Cols	199	4	0	1	0	8	7517	2.78×10^{-3}	5.46×10^{-3}	+0.245
Gurobi	0.1%	SG-Ter-Mer	SA-Aug	62	3	1	2	0	1	456	2.63×10^{-4}	1.58×10^{-3}	−0.533
Gurobi	0.1%	SG-Ter-Mer	SA-Mat	64	3	0	0	1	3	341	2.95×10^{-6}	3.53×10^{-5}	−0.672
Gurobi	0.1%	SG-Ter-Mer	Ruiz	64	3	1	0	0	3	889	0.313	0.341	−0.145
Gurobi	0.1%	SG-Ter-Mer	Ruiz+Cols	64	3	1	0	0	6	599	3.19×10^{-3}	5.46×10^{-3}	+0.424
Gurobi	0.1%	Full	SA-Aug	28	2	3	1	0	2	124	0.0735	0.110	−0.389
Gurobi	0.1%	Full	SA-Mat	27	2	4	1	0	0	173	0.714	0.714	−0.085
Gurobi	0.1%	Full	Ruiz	28	2	3	0	1	3	144	0.186	0.248	+0.291
Gurobi	0.1%	Full	Ruiz+Cols	28	2	4	0	0	3	149	0.227	0.272	−0.266
CPLEX	1%	Simple	SA-Aug	204	0	0	0	0	0	9377	0.202	0.346	−0.103
CPLEX	1%	Simple	SA-Mat	204	0	0	0	0	0	7460	3.89×10^{-4}	2.33×10^{-3}	+0.286
CPLEX	1%	Simple	Ruiz	204	0	0	0	0	0	9496	0.256	0.384	+0.092
CPLEX	1%	Simple	Ruiz+Cols	204	0	0	0	0	0	8958	0.0762	0.183	+0.143
CPLEX	1%	SG-Ter-Mer	SA-Aug	68	0	0	0	0	0	1068	0.521	0.521	+0.090
CPLEX	1%	SG-Ter-Mer	SA-Mat	68	0	0	0	0	0	740	8.15×10^{-3}	0.0326	+0.369

Continued on next page

Table S6 (continued)

Solver	Gap	Class	Variant	n	UM	D_{TO}	D_{A}	D_{mix}	C_1	W	p	q	r_{rb}
CPLEX	1%	SG-Ter-Mer	Ruiz	68	0	0	0	0	0	1041	0.420	0.458	+0.113
CPLEX	1%	SG-Ter-Mer	Ruiz+Cols	68	0	0	0	0	0	438	7.09×10^{-6}	8.50×10^{-5}	+0.627
CPLEX	1%	Full	SA-Aug	29	0	4	0	1	1	172	0.336	0.428	+0.209
CPLEX	1%	Full	SA-Mat	28	0	5	0	1	0	131	0.104	0.208	−0.355
CPLEX	1%	Full	Ruiz	31	0	2	0	1	3	200	0.357	0.428	−0.194
CPLEX	1%	Full	Ruiz+Cols	30	0	3	0	1	2	130	0.0345	0.104	+0.441
CPLEX	0.1%	Simple	SA-Aug	204	0	0	0	0	2	8765	0.0453	0.121	−0.162
CPLEX	0.1%	Simple	SA-Mat	203	0	1	0	0	0	10056	0.723	0.723	−0.029
CPLEX	0.1%	Simple	Ruiz	204	0	0	0	0	2	9364	0.196	0.356	−0.104
CPLEX	0.1%	Simple	Ruiz+Cols	203	0	1	0	0	2	8092	6.98×10^{-3}	0.0419	+0.218
CPLEX	0.1%	SG-Ter-Mer	SA-Aug	68	0	0	0	0	0	1085	0.591	0.709	+0.075
CPLEX	0.1%	SG-Ter-Mer	SA-Mat	68	0	0	0	0	0	985	0.251	0.372	−0.160
CPLEX	0.1%	SG-Ter-Mer	Ruiz	68	0	0	0	0	0	853	0.0505	0.121	−0.273
CPLEX	0.1%	SG-Ter-Mer	Ruiz+Cols	68	0	0	0	0	1	473	1.89×10^{-5}	2.27×10^{-4}	+0.597
CPLEX	0.1%	Full	SA-Aug	27	0	6	0	1	3	173	0.714	0.723	−0.085
CPLEX	0.1%	Full	SA-Mat	26	0	6	0	2	4	125	0.208	0.356	−0.288
CPLEX	0.1%	Full	Ruiz	30	0	3	0	1	6	134	0.0427	0.121	−0.424
CPLEX	0.1%	Full	Ruiz+Cols	27	0	6	0	1	6	143	0.279	0.372	+0.243

One-sided-abort sensitivity. Replacing the primary $T_{\max}$ convention by $10T_{\max}$ for a one-sided algorithmic/pipeline abort leaves every effect direction and inferential decision unchanged. The largest change among the displayed Gurobi structure-aware rows is Full/SA-Aug at 0.1%: $W = 124 \to 125$, $p = 0.07354 \to 0.07742$, $q_{\mathrm{BH}} = 0.11031 \to 0.11612$, and $r_{\mathrm{rb}} = -0.389 \to -0.384$. All other Gurobi structure-aware rows are unchanged; CPLEX structure-aware W, p, and r_{rb} are unchanged, with only minor BH shifts caused by other members of the twelve-test family. The sensitivity is reproduced by `scripts/paired_tests.py --abort-penalty par10`.

Cluster-bootstrap accounting for the conditioning comparison. At 1%, 298 of 306 instances have at least one complete conditioning–speedup pair; eight are lost because Base errored. The whole-instance bootstrap uses 1185 complete variant–instance observations (seven of the possible 298×4 lack the conditioning diagnostic). At 0.1%, 288 instances contribute; eighteen Base errors remove nine Simple, six SG-Ter-Mer, and three Full instances. Of 1152 potential rows, 1135 enter the joint comparison: twelve lack a usable variant speedup and five more lack the solver-conditioning diagnostic. Each of the 2000 resamples keeps all available formulation rows of an instance together. At 1%, the within-class solver-level coefficients are −0.31 (Simple), −0.13 (SG-Ter-Mer), and −0.12 (Full); their varying magnitudes reinforce that the pooled coefficient is descriptive rather than an instance-level predictor.

Exploratory dependence check. If repeated rows and seeds are naively treated as independent observations, the first-order partial rank correlation between runtime improvement and the solver-facing conditioning diagnostic is approximately −0.25 after controlling for loaded nonzeros, and a Williams–Steiger comparison gives $Z \approx 6.1$. Because both calculations ignore the clustered generator design, they are reported only as descriptive diagnostics and support no confirmatory claim.

Table S7: Available solver-internal-scaling sweep over both solvers, both MIPGap settings, and all three classes. Each cell is solver-time sgm ($s = 1$ s)/PAR10$_{\text{avail}}$ in seconds, using the descriptive revision-campaign convention (timeouts receive 36000 s; pure pipeline errors are excluded). SA-Aug and SA-Mat retain the solver default. For Gurobi, Off/A/B/C mean `ScaleFlag`=0/1/2/3. For CPLEX they mean `Read::Scale`=−1/0/1/not applicable. The CPLEX Full 0.1% Scale=0 artifact is partial (26 of 34 rows), and Scale=1 is absent because the four-day sweep budget expired; all other configuration folders contain the manifest-declared number of rows. The lower panel audits the apparently fast CPLEX Scale=−1 Full cell directly from the stored original-scale verifier fields.

Panel A: timing sweep (sgm/PAR10$_{\text{avail}}$)

Solver	Gap	Class	Base	SA-Aug	SA-Mat	Off	A	B	C
Gurobi	1%	Simple	3.936/11.26	2.133/2.50	2.166/2.56	41.969/1541.92	5.811/20.39	4.402/12.60	3.791/10.79
Gurobi	1%	SG-Ter-Mer	14.087/564.33	8.282/554.10	8.635/554.64	56.625/740.17	14.664/566.35	20.907/589.78	21.475/580.46
Gurobi	1%	Full	109.375/3448.17	95.239/3349.65	102.025/3458.10	561.945/9716.08	128.449/2444.67	138.112/2497.10	133.828/3468.97
Gurobi	0.1%	Simple	9.656/36.16	6.419/210.83	6.248/202.17	89.358/2031.10	15.299/219.91	10.607/33.11	9.438/35.10
Gurobi	0.1%	SG-Ter-Mer	14.587/566.63	8.811/554.42	9.066/555.19	54.784/727.17	16.972/620.44	11.753/32.85	23.102/585.62
Gurobi	0.1%	Full	437.243/4766.52	388.446/4714.38	456.365/4821.25	396.978/9577.84	337.331/4621.00	347.613/3687.08	353.389/4697.74
CPLEX	1%	Simple	2.621/7.24	1.725/3.65	1.861/4.01	2.500/3.93	2.549/7.59	2.431/7.18	–
CPLEX	1%	SG-Ter-Mer	2.253/3.57	2.400/10.26	2.309/5.94	5.168/12.29	2.229/3.53	2.175/5.66	–
CPLEX	1%	Full	147.821/5609.27	127.360/3616.84	120.881/6509.40	14.381/35.45	159.012/5621.84	189.129/5644.99	–
CPLEX	0.1%	Simple	5.143/215.88	4.266/188.91	4.450/203.68	7.634/209.70	5.203/216.59	5.625/369.68	–
CPLEX	0.1%	SG-Ter-Mer	2.664/22.29	2.678/24.19	2.667/19.50	5.844/13.69	2.725/23.13	2.477/11.78	–
CPLEX	0.1%	Full	425.043/9419.41	426.712/9179.82	387.204/11235.11	13.476/24.97	303.376/6196.93†	–	–

Panel B: CPLEX Full, `Read::Scale`= −1, original-scale audit

Gap	Runs	Strict pass	Mixed pass	max v_{row}	max v_{int}	max $\lvert z_{\text{solver}} - z_{\text{recomputed}} \rvert$	Strict-fail identifiers
1%	34	26	26	1.3149×10^{-2}	0	3.73×10^{-8}	`002,003,004,012,014,022,023,044`
0.1%	34	26	26	1.3149×10^{-2}	0	3.73×10^{-8}	`002,003,004,012,014,022,023,044`

Notes. The raw feasibility, integrality, and recomputed-objective fields are present for all 34 runs; the two gap folders contain identical objectives and verifier values. At 1%, 11 of 34 Scale=−1 objectives differ from the default-base objective by more than twice the stopping tolerance; six of those 11 also fail strict verification, while five pass it. Thus the speedup is real as timing but cannot be classified as a reliable solution improvement. The campaign's generic `mip_equivalente` field labels every variant named `base` as `BASE`, irrespective of its internal-scaling parameter, so the pass counts above are rebuilt from the raw verifier fields rather than that label. †Partial artifact; its descriptive value is not comparable to a complete 34-instance cell.

Table S8: Archived runtime comparison for the operational coupling-stage activation campaign (Gurobi, 1% MIPGap). Solver-time shifted geometric mean (sgm, $s = 1$ s) and descriptive $\mathrm{PAR10}_{\mathrm{avail}}$ are seconds. The accompanying activation diagnostic found zero coupling-stage activations and byte-identical exports for each full/local-only pair; their small runtime differences therefore measure run-to-run variability, not a coupling-stage effect. The coupling-only variant coincides with Base at the exported-model level.

Class	Variant	sgm (s)	$\mathrm{PAR10}_{\mathrm{avail}}$ (s)
Simple	Base	3.73	10.47
	SA-Aug (local+coupling)	2.02	2.42
	SA-Aug, local only	2.01	2.38
	SA-Aug, coupling only	3.74	10.44
	SA-Mat (local+coupling)	2.04	2.43
	SA-Mat, local only	2.06	2.48
	SA-Mat, coupling only	3.73	10.26
Full	Base	82.55	3379.7
	SA-Aug (local+coupling)	71.41	3315.9
	SA-Aug, local only	70.94	3314.4
	SA-Aug, coupling only	83.69	3381.5
	SA-Mat (local+coupling)	78.75	4384.9
	SA-Mat, local only	78.88	4386.1
	SA-Mat, coupling only	90.21	3394.1
SG-Ter-Mer	Base	14.42	565.6
	SA-Aug (local+coupling)	8.55	555.1
	SA-Aug, local only	8.58	555.2
	SA-Aug, coupling only	14.45	565.9
	SA-Mat (local+coupling)	8.96	555.6
	SA-Mat, local only	8.83	555.6
	SA-Mat, coupling only	14.43	565.8

In SG-Ter-Mer one near-limit instance times out under every variant in this independent batch, so $\mathrm{PAR10}_{\mathrm{avail}}$ is dominated by a single 36000-s penalty. In Full, three or four runs time out per variant, so the raw sgm values use different solved subsets and are descriptive. The attribution is the export/activation result: full equals local only, and coupling only equals Base.

Table S9: Direct SA–Flat reliability audit under Gurobi. Each row uses all available records from one dedicated class–tolerance batch; both SA and Flat were rerun in that batch. Timeouts (TO), strict/mixed original-scale verification failures (S/M), and fixed-pool PAR10 are reported for both coverage policies. Each PAR10 cell is ITT/strict/mixed in seconds, on the capped solver-time cost $c^{(k,\text{solver})}$ of the main text; recomputing with the preprocessing-plus-solver cost shifts every cell by less than 0.2 s and changes no ordering.

Gap	Class	Kernel	N	TO SA/Flat	SA fail S/M	Flat fail S/M	SA PAR10 I/S/M	Flat PAR10 I/S/M
1%	Simple	Augmented	204	0/0	12/12	13/12	2.4/2119.8/2119.8	2.4/2296.2/2119.7
	Simple	Matrix-only	204	0/0	19/18	19/18	2.4/3355.1/3178.7	2.4/3355.1/3178.6
	SG-Ter-Mer	Augmented	68	1/1	1/0	2/0	554.5/1083.9/554.5	541.1/1598.9/541.1
	SG-Ter-Mer	Matrix-only	68	1/1	1/0	1/0	555.0/1083.7/555.0	551.6/1080.8/551.6
	Full	Augmented	34	3/3	6/6	3/3	3312.1/9643.7/9643.7	3374.3/6464.9/6464.9
	Full	Matrix-only	34	3/3	2/2	4/4	3420.9/5533.6/5533.6	3430.6/7554.0/7554.0
0.1%	Simple	Augmented	204	1/0	22/18	25/22	211.9/4090.0/3386.8	30.3/4436.5/3908.1
	Simple	Matrix-only	204	1/1	30/26	30/26	203.5/5491.9/4787.2	203.7/5492.1/4787.4
	SG-Ter-Mer	Augmented	68	1/1	1/0	1/0	556.0/1085.4/556.0	542.4/1071.4/542.4
	SG-Ter-Mer	Matrix-only	68	1/1	1/0	1/0	556.5/1085.6/556.5	552.8/1082.0/552.8
	Full	Augmented	30	3/3	2/2	2/1	3951.0/6342.2/6342.2	4004.3/6290.8/5109.3
	Full	Matrix-only	30	4/4	2/1	0/0	5193.8/7585.0/6389.5	5154.8/5154.8/5154.8

Notes. There are no algorithmic or pipeline aborts in these direct batches. Strict penalizes every strict original-scale failure; mixed penalizes only a failure that also lacks a reliable counterpart on the same instance, as defined in the main text. At Full 0.1%, instances 045–048 are absent from both policies. Under an equal-penalty imputation for those four unrecorded instances the ordering is unchanged and the added pairs are operational ties; that assumption does not address selection under unequal missing outcomes, for which the adversarial signed-rank sensitivity is reported in the main text and in the Notes to Tables S10–S12. The key reliability sensitivity is visible in Simple/Augmented: the small Flat solver-time advantage at 0.1% reverses under both strict and mixed costs. SG-Ter-Mer/Augmented retains a small descriptive Flat advantage because neither policy has a mixed failure.

Table S10: Magnitude and dependence sensitivities for the direct SA–Flat contrasts (exploratory endpoint hierarchy; see the main text). Panel A reports instance-level and operational-family inference for the two central 0.1% augmented contrasts on the prototype preprocessing-plus-solver implementation endpoint and on the capped solver-time decomposition, on the shared paired variable $d_i = \log(T_{\mathrm{Flat,i}}/T_{\mathrm{SA,i}})$, with the Hodges–Lehmann pseudomedian expressed as the typical SA/Flat ratio $\exp(-\widehat{\theta}_{\mathrm{HL}})$. Panel B reports the corresponding Gurobi Work-unit sensitivity. Panel C reports the own-incumbent path-specific diagnostic, which does not hold the integer solution fixed across policies.

Panel A: paired log time ratio and dependence (central augmented 0.1% contrasts)

Class	Endpoint	n/G	med. SA/Flat	HL SA/Flat	family-bootstrap 95% CI	p_{inst}	q_6	q_{12}	p_{block}	$q_{6,\mathrm{block}}$	$q_{12,\mathrm{block}}$
Simple	prototype pre+solve	204/34	1.0320	1.0359	[1.0274, 1.0418]	3.3×10^{-11}	2.0×10^{-10}	1.3×10^{-10}	5.1×10^{-8}	3.1×10^{-7}	2.1×10^{-7}
Simple	solver	204/34	1.0093	1.0135	[1.0019, 1.0212]	0.00173	0.01038	0.02075	0.00266	0.01595	0.03190
SG-Ter-Mer	prototype pre+solve	67/34	1.0473	1.1577	[1.0170, 1.2065]	0.00031	0.00063	0.00063	0.00056	0.00111	0.00117
SG-Ter-Mer	solver	67/34	1.0159	1.1229	[0.9797, 1.1309]	0.02616	0.05232	0.09346	0.03320	0.06640	0.11980

Panel A2: prototype preprocessing-plus-solver endpoint, all twelve contrasts

Adjusted values only; effect directions and median SA/Flat ratios are reported in Table 18 of the main text.

Gap	Class	Kernel	n	q_6	q_{12}	$q_{6,\mathrm{block}}$	$q_{12,\mathrm{block}}$
1%	Simple	Augmented	204	7.9×10^{-26}	1.6×10^{-25}	1.0×10^{-9}	2.1×10^{-9}
	Simple	Matrix-only	204	3.0×10^{-32}	6.0×10^{-32}	7.0×10^{-10}	1.4×10^{-9}
	SG-Ter-Mer	Augmented	67	0.00057	0.00063	0.00117	0.00117
	SG-Ter-Mer	Matrix-only	67	0.01126	0.01287	0.01893	0.02163
	Full	Augmented	31	0.1502	0.1502	0.1502	0.1502
	Full	Matrix-only	32	0.3403	0.3712	0.3403	0.3712
0.1%	Simple	Augmented	204	2.0×10^{-10}	1.3×10^{-10}	3.1×10^{-7}	2.1×10^{-7}
	Simple	Matrix-only	203	4.9×10^{-8}	4.9×10^{-8}	0.00042	0.00042
	SG-Ter-Mer	Augmented	67	0.00063	0.00063	0.00111	0.00117
	SG-Ter-Mer	Matrix-only	67	0.02239	0.02468	0.02765	0.03072
	Full	Augmented	27	0.02239	0.02487	0.02765	0.02798
	Full	Matrix-only	27	0.8408	0.8408	0.8408	0.8408

Panel B: deterministic effort ($\mathrm{Work}_{\mathrm{Flat}} - \mathrm{Work}_{\mathrm{SA}}$)

Class	n	Exact zeros	med. Work SA	med. Work Flat	r_{rb}	p_{block}	$q_{6,\mathrm{block}}$	$q_{12,\mathrm{block}}$
Simple	204	138	2	2	−0.382	0.01196	0.03587	0.04782
SG-Ter-Mer	67	31	7	4	−0.649	0.000970	0.00582	0.00637

Panel C: median own-incumbent path-diagnostic ratio

Gap	Class	Kernel	n	med. $\rho^{\mathrm{solver}}_{\kappa,\mathrm{SA}}/\rho^{\mathrm{solver}}_{\kappa,\mathrm{Flat}}$
1%	Simple	Augmented	204	1.000
	Simple	Matrix-only	204	1.000
	SG-Ter-Mer	Augmented	67	0.626
	SG-Ter-Mer	Matrix-only	67	0.849
	Full	Augmented	31	0.628
	Full	Matrix-only	32	0.849
0.1%	Simple	Augmented	204	1.000
	Simple	Matrix-only	203	1.000
	SG-Ter-Mer	Augmented	67	0.629
	SG-Ter-Mer	Matrix-only	67	0.849
	Full	Augmented	27	0.634

Table S11: Attrition, original-batch preprocessing timing, and release-versus-diagnostic cached-metadata sensitivities for the direct SA–Flat controls. Panel D reports single-call preprocessing summaries from the original fixed-order solve batches. Panels E1–E2 use per-instance median preprocessing times estimated from the replicated low-contention protocols; uncertainty in those estimated medians is not propagated into the derived inferential quantities.

Panel D: per-contrast attrition and preprocessing-time summaries

Gap	Class	Kernel	Nominal	Recorded	Common pool	Double noncompl.	n used	med. $T_{\text{pre}}^{\text{SA}}$ (s)	med. $T_{\text{pre}}^{\text{Flat}}$ (s)	med. diff (s)
1%	Simple	Augmented	204	204	204	0	204	0.074	0.025	+0.050
	Simple	Matrix-only	204	204	204	0	204	0.077	0.025	+0.053
	SG-Ter-Mer	Augmented	68	68	68	1	67	0.049	0.019	+0.031
	SG-Ter-Mer	Matrix-only	68	68	68	1	67	0.051	0.019	+0.033
	Full	Augmented	34	34	34	3	31	0.121	0.041	+0.080
	Full	Matrix-only	34	34	34	2	32	0.125	0.041	+0.084
0.1%	Simple	Augmented	204	204	204	0	204	0.077	0.025	+0.051
	Simple	Matrix-only	204	204	204	1	203	0.079	0.025	+0.054
	SG-Ter-Mer	Augmented	68	68	68	1	67	0.051	0.019	+0.032
	SG-Ter-Mer	Matrix-only	68	68	68	1	67	0.052	0.019	+0.033
	Full	Augmented	34	30	30	3	27	0.120	0.041	+0.079
	Full	Matrix-only	34	30	30	3	27	0.124	0.041	+0.083

Panel E1: release-build cached-metadata endpoint (implementation-cost sensitivity of record)

Gap	Class	Kernel	n	med. SA/Flat	r_{rb}	q_6	q_{12}	$q_{6,\text{block}}$	$q_{12,\text{block}}$
1%	Simple	Aug	204	1.0004	−0.090	0.318	0.364	0.337	0.450
	Simple	Mat	204	1.0011	−0.167	0.115	0.092	0.173	0.162
	SG-Ter-Mer	Aug	67	1.0075	−0.303	0.115	0.092	0.173	0.120
	SG-Ter-Mer	Mat	67	0.9976	−0.140	0.318	0.381	0.337	0.450
	Full†	Aug	31	0.9193	+0.355	0.173	0.173	0.173	0.173
	Full†	Mat	32	0.9866	+0.246	0.318	0.364	0.337	0.396
0.1%	Simple	Aug	204	1.0094	−0.253	0.010	0.021	0.016	0.031
	Simple	Mat	203	0.9981	+0.089	0.410	0.364	0.546	0.469
	SG-Ter-Mer	Aug	67	1.0159	−0.313	0.051	0.092	0.065	0.120
	SG-Ter-Mer	Mat	67	1.0008	−0.109	0.526	0.478	0.546	0.496
	Full†	Aug	27	0.8771	+0.529	0.045	0.090	0.045	0.090
	Full†	Mat	27	0.9803	+0.053	0.822	0.822	0.822	0.822

† Approximate cross-build sensitivity; five Full instances exhibit last-bit model-construction differences. See Section S2.1.

Panel E2: non-optimized diagnostic-build audit

Gap	Class	Kernel	n	med. SA/Flat	r_{rb}	q_6	q_{12}	$q_{6,\text{block}}$	$q_{12,\text{block}}$
1%	Simple	Aug	204	1.0008	−0.124	0.185	0.212	0.258	0.295
	Simple	Mat	204	1.0019	−0.333	0.0002	0.0005	0.004	0.008
	SG-Ter-Mer	Aug	67	1.0074	−0.303	0.093	0.075	0.119	0.095
	SG-Ter-Mer	Mat	67	0.9976	−0.148	0.291	0.388	0.310	0.414
	Full	Aug	31	0.9197	+0.355	0.173	0.173	0.173	0.173
	Full	Mat	32	0.9869	+0.242	0.286	0.358	0.286	0.358
0.1%	Simple	Aug	204	1.0096	−0.259	0.008	0.008	0.013	0.013
	Simple	Mat	203	0.9983	+0.065	0.506	0.460	0.633	0.576
	SG-Ter-Mer	Aug	67	1.0159	−0.314	0.051	0.075	0.065	0.095
	SG-Ter-Mer	Mat	67	1.0036	−0.116	0.506	0.460	0.633	0.512
	Full	Aug	27	0.8771	+0.529	0.045	0.060	0.045	0.060
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Section S2.1. Timing, build, and transfer-factor audits

Table S12: timing, build, and cross-environment transfer-factor audits for the direct SA–Flat controls. Panel E3 reports the number and identity of fast-class contrasts with $q_6 < 0.05$ over the hypothetical transfer-factor grids; Panel E4 reports the corresponding Full-Augmented sensitivity; Panel E5 reports the release-build absolute preprocessing times. The accompanying text documents the build comparison and the export-invariance audit. The transfer factors are hypothetical: the workstation-to-ClusterUY transfer was never observed, and the only measured perturbation is the change of toolchain on the same machine.

† Five Full instances exhibit last-bit model-construction differences, so the Full release-cached values in Panel E1 should be read as approximate build sensitivities rather than exact recombinations of one model; see the build-audit details below and `results-revision/preproc-timing/build_invariance.csv`. On the cached counterpart, the role-blind rule has the lower per-instance median in 585/612 records under the pinned release build, 607/612 under the pinned diagnostic build and 408/612 under the concurrent diagnostic protocol. For a descriptive within-instance dispersion comparison, we call record i a *within-dispersion tie* for build b when

$$\left|\widetilde{T}^{\text{cached},b}_{\text{preproc,SA},i} - \widetilde{T}^{\text{cached},b}_{\text{preproc,Flat},i}\right| \le B^b_i, \qquad B^b_i = \max\left\{\mathrm{IQR}\left(T^{\text{cached},b}_{\text{preproc,SA},i}\right), \mathrm{IQR}\left(T^{\text{cached},b}_{\text{preproc,Flat},i}\right)\right\},$$

where $\widetilde{T}$ denotes the median and the interquartile ranges are taken over the $R = 20$ repetitions; the quantities are cached *preprocessing* times, not the preprocessing-plus-solver endpoint. Using this descriptive scale-of-dispersion classification, the three-way classification (role-blind-favoured, selective-favoured, tie) is 461/1/150 under the pinned release build, 554/0/58 under the pinned diagnostic build and 37/0/575 under the concurrent protocol; each row sums to 612. The band is an ad hoc scale-of-dispersion heuristic based on the larger marginal IQR; it is not an uncertainty estimate for the paired median difference, nor a confidence interval, an equivalence test or a domain-calibrated practical-significance margin, so records inside it are within-dispersion ties rather than statistically indeterminate. A sharper screen would use the per-repetition paired differences D^b_{ir} and their IQR_r, which the archived summaries do not retain. These counts condition on the $R = 20$ per-instance median estimates and do not propagate within-instance timing uncertainty. The three record-level counts are computed by `scripts/preproc_timing_replication.py` from the `cached_sa_s` and `cached_flat_s` columns of `results-revision/preproc-timing/replicated_pinned_release.csv` (pinned release), `results-revision/preproc-timing/replicated_pinned.csv` (pinned diagnostic) and `results-revision/preproc-timing/replicated.csv` (concurrent diagnostic).

Panel E3: release-build cross-environment transfer-factor sensitivity

Grid point	Fast-class contrasts with $q_6 < 0.05$ (of eight)					
Common λ	0.25	0.5	1	2	4	8
	1	1	1	1	2	2
Ratio $\rho = \lambda_{\text{SA}}/\lambda_{\text{Flat}}$	0.25	0.5	1	2	4	—
	1	1	1	3	5	—

Measurement and build-audit details.

The crossing contrasts are identified as follows. On the common-factor grid, Simple/Augmented at 0.1% crosses at every point; Simple/Matrix-only at 1% joins it at $\lambda = 4$ and 8. On the policy-specific grid, Simple/Augmented at 0.1% crosses at every point; at $\rho = 2$ it is joined by SG-Ter-Mer/Augmented at 0.1% and Simple/Matrix-only at 1%; at $\rho = 4$ also by SG-Ter-Mer/Augmented at 1% and Simple/Augmented at 1%. Panel E3 is restricted to the eight fast-class contrasts. In the separate Full-Augmented 0.1% calculation, $q_6 = 0.045$ at every evaluated common-factor point and at the policy-specific points $\rho \le 2$, falling to 0.036 at $\rho = 4$; the joint value q_{12} remains above 0.05 except at $\rho = 4$, where it is 0.042 (Panel E4). None of these nominal Full crossings is retained by the adversarial missing-pair sensitivity. The $q_{12} = 0.042$ crossing at the extreme evaluated point $\rho = 4$ is moreover conditional on a hypothetical policy-specific transfer factor, on approximate cross-build Full models, and on the 27 observed eligible pairs; it is not treated as evidence of a robust Full coverage ordering. No other Full contrast has $q_6 < 0.05$ anywhere on the grids.

Panel E4: Full-Augmented at 0.1% across the transfer grids (approximate cross-build row; $n = 27$).

Common λ	0.25	0.5	1	2	4	8
q_6	0.045	0.045	0.045	0.045	0.045	0.045
q_{12}	0.090	0.090	0.090	0.063	0.060	0.060
Ratio ρ	**0.25**	**0.5**	**1**	**2**	**4**	**—**
q_6	0.045	0.045	0.045	0.045	0.036	—
q_{12}	0.090	0.090	0.090	0.060	**0.042**	—

The adversarial missing-pair sensitivity was recomputed at every grid point on this cached endpoint: its effective nonzero sample remains $n = 31$ and $p_{\text{adv}} = 0.433$ at all eleven points, so no point falls below 0.05 (the corresponding value on the prototype endpoint is 0.47). For the observed-pair, non-adversarial analysis, the unadjusted values are likewise invariant along both grids ($p = 0.0151$, $p_{\text{block}} = 0.0151$), because scaling the preprocessing term cannot change the paired ranks here—that term is milliseconds against solver times of seconds—so only the BH-adjusted values move, and they move because the other contrasts in the family do. Because every Full instance is a singleton operational family, the family-level values coincide with the instance-level values at every grid point: $q_{6,\text{block}} = q_6$ and $q_{12,\text{block}} = q_{12}$.

The release-build endpoint (Panel E1) is the implementation-cost sensitivity used in the main text; the diagnostic-build results (Panel E2) are retained only to show the sensitivity of the cached comparison to compiler optimization. Panel E3 reports within-tolerance q_6 crossings only; these counts do not imply retention under the joint or operational-family adjustments. Among the eight fast-class contrasts summarized in Panel E3, Simple-Augmented at 0.1% is the only contrast with $q_6 < 0.05$ at every evaluated grid point; the separate approximate Full-Augmented row also has $q_6 < 0.05$ throughout, but its evidence is conditional on the observed eligible pairs and is not retained by the adversarial missing-pair sensitivity. Additional within-tolerance q_6 crossings first appear at the evaluated point $\rho = 2$; the exact transition threshold between 1 and

2 is not identified by this grid. The transfer factors are hypothetical: the workstation-to-ClusterUY transfer was never observed, and the only measured perturbation is the change of toolchain on the same machine. Comparing Panels E1 and E2 gives the implied toolchain ratio. Writing, for build $b \in \{\text{release}, \text{diagnostic}\}$, class c and kernel k,

$$R_{b,c,k} = \exp\Big[\text{median}_i \log\big(T^{\text{cached},b}_{\text{preproc,SA},i} / T^{\text{cached},b}_{\text{preproc,Flat},i}\big)\Big], \qquad \rho^{\text{toolchain}}_{c,k} = \frac{R_{\text{release},c,k}}{R_{\text{diagnostic},c,k}},$$

The quantities in this definition are cached *preprocessing* times only, not the cached preprocessing-plus-solver endpoint of the main-text table; it is a ratio of exponentiated median log-ratios, not a ratio of marginal median preprocessing times (the quantity computed by `scripts/cached_transfer_sensitivity.py`). Its values are 0.947, 0.942, 1.012, 0.994, 0.978 and 0.966 for Full/Aug, Full/Mat, SG-Ter-Mer/Aug, SG-Ter-Mer/Mat, Simple/Aug and Simple/Mat; optimization leaves the prototype preprocessing-call ratios essentially unchanged (5.38–6.12× against 5.07–6.00×) and the metadata share at 81–83%. Exporting the preprocessed model with both builds over all 306 instances, both kernels and both coverage policies gives 1204/1224 byte-identical exports (`scripts/build_invariance_audit.py`; full record in `results-revision/preproc-timing/build_invariance.csv`). All Simple and SG-Ter-Mer records are identical. The 20 exceptions are five Full instances under all four variants, where the nonlinear-hydro linearization coefficients differ by one unit in the last place (4.7×10^{-16} relative); the audit localizes the discrepancy to optimization-dependent floating-point evaluation during model construction rather than to a change in the selected scaling factors or safeguard decisions. The complete contrast-by-grid values—every contrast, grid point, median ratio, r_{rb}, p, q_6 and q_{12}—are in `results-revision/preproc-timing/cached_transfer_sensitivity.csv` of the reproducibility package, regenerated by `scripts/cached_transfer_sensitivity.py --dump-csv`.

Panel E5: release-build absolute preprocessing times, in milliseconds as median [IQR]; an IQR below 0.001 ms is shown as < 0.001 rather than rounded to zero. The absolute times quoted elsewhere in these notes are from the diagnostic build.

		Prototype preprocessing call (ms)		Cached preprocessing counterpart (ms)	
Class	Kernel	SA	Flat	SA	Flat
Simple	Augmented	8.447 [0.093]	1.410 [0.044]	1.362 [0.012]	1.288 [0.043]
Simple	Matrix-only	8.606 [0.100]	1.405 [0.048]	1.528 [0.011]	1.283 [0.044]
SG-Ter-Mer	Augmented	5.335 [0.041]	0.991 [0.005]	0.911 [0.006]	0.894 [0.004]
SG-Ter-Mer	Matrix-only	5.433 [0.040]	0.975 [0.006]	1.006 [0.009]	0.877 [0.006]
Full	Augmented	13.607 [0.121]	2.401 [0.119]	2.248 [0.044]	2.220 [0.121]
Full	Matrix-only	13.878 [0.147]	2.384 [0.085]	2.514 [0.093]	2.203 [0.077]

Notes to Tables S10–S12. Both time endpoints use the paired variable $d_i = \log(T_{\text{Flat,i}}/T_{\text{SA,i}})$; negative values favor Flat. The test, the effect size, and the median ratio are computed on this one scale and share its direction convention but do not target an identical estimand; the Hodges–Lehmann column is the location estimate aligned with the signed-rank test. G is the number of operational families. Simple groups the six `casoXX[a--f]` scenarios with a common root; SG-Ter-Mer groups each `instanceXXX/instance(XXX+48)` pair (34 pairs, no singletons); Full is singleton by construction. The exact block-sign test preserves each scenario's absolute-rank contribution, sums those contributions within family, and flips one common sign per family; its null distribution is computed by dynamic programming. The percentile bootstrap resamples whole families ($B = 2000$, seed 12345; the pre+solve and solver-time intervals are drawn from one RNG stream in that order). Negative Work r_{rb} favors Flat under the sign convention of the main table. Work reproduces both augmented 0.1% solver-time directions but, with medians 2 versus 2 in Simple and 7 versus 4 in SG-Ter-Mer, does not turn the small solver-time effects into large speedups; a single default-seed run per cell means seed- and trajectory-level variability is not eliminated. In Panel C each policy fixes the integer variables to the incumbent found by that same run. The ratio can therefore reflect both representation and a different integer solution; no common-incumbent diagnostic can be reconstructed from the archived artifacts. In Panel D, "Recorded" counts instances present in the batch CSV (at Full 0.1%, instances 045–048 were not recorded), the common pool removes uniformly unavailable instances (none here beyond the unrecorded ones), and double noncompletions are pairs in which neither policy completes. No pair is dropped for a missing time and no exact-zero difference occurs; n used applies to both endpoints. In these batches the preprocessing calls are timed once each, with the C++ high-resolution wall clock, in the batch's fixed (un-randomized) execution order and without repetitions; per-policy IQRs are at most 0.003 s in every cell. Timing environment for the preprocessing replication: Intel Core i7-13620H (10 physical / 16 logical cores, 0.4–4.9 GHz, `powersave` governor), Ubuntu 22.04.5 with Linux 6.8.0, g++ 11.4.0 at `-std=c++17 -g` (no `-O` flag), binary `code/build_diag/SistemaElectrico`; the pinned protocol binds one worker to a single logical CPU with `taskset`, the concurrent protocol uses 12 unpinned workers. The release-build replication uses the same source commit, hardware, operating system, compiler, libraries, timing protocol, seeds, execution order and affinity policy, compiled with `-std=c++17 -O2 -DNDEBUG` into `code/build_release/SistemaElectrico`; the two builds differ only in optimization flags. The solve batches ran on ClusterUY as described in the main text, so the cached-metadata endpoint is a cross-protocol counterfactual.

Order-randomized preprocessing replication (`scripts/preproc_timing_replication.py`; $R = 20$, seeded order per repetition, fresh process per call, MIP never solved). Two protocols were run: a *concurrent* stress test (12 workers on a 16-core workstation, no CPU pinning) and a *low-contention* estimate (one worker, every call pinned to one logical CPU via `--pin-cpu`). The pinned protocol is the cleaner estimator—relative IQR 0.00–0.02 against 0.18–1.83—and gives, per class and kernel, selective versus role-blind medians and paired ratios: Simple/Aug 0.0487 vs 0.0083 (5.85×; cached 0.0087 vs 0.0081, 1.08×); Simple/Mat 0.0499 vs 0.0083 (6.00×; cached 0.0099 vs 0.0081, 1.23×); SG-Ter-Mer/Aug 0.0300 vs 0.0059 (5.07×; cached 0.0058 vs 0.0057, 1.01×); SG-Ter-Mer/Mat 0.0307 vs 0.0058 (5.28×; cached 0.0065 vs 0.0056, 1.15×); Full/Aug 0.0773 vs 0.0136 (5.71×; cached 0.0141 vs 0.0132, 1.07×); Full/Mat 0.0797 vs 0.0137 (5.81×; cached 0.0161 vs 0.0133, 1.21×). All 612 instance–kernel records favor the role-blind rule on the full call under both protocols. On the cached counterpart the role-blind rule has the lower median in 20–32 of the 34 operational families per cell, so that residual is descriptive rather than uniform.

The concurrent protocol is retained as a stress test of the direction under adverse measurement conditions. Its per-class and per-kernel values—medians of the per-instance median call time (selective versus role-blind, seconds), the median paired ratio, the cached-metadata counterpart $T_{\text{diagnosis}} + T_{\text{application}}$, and the relative interquartile range, which replaces the coefficient of variation because the latter is unstable for short, skewed timings—are: Simple/Aug 0.272 vs 0.031 (8.38×; cached 0.032 vs 0.029, 1.08×; relative IQR 0.32 vs 1.51); Simple/Mat 0.281 vs 0.031 (8.58×; cached 0.040 vs 0.029, 1.40×; 0.32 vs 1.58); SG-Ter-Mer/Aug 0.165 vs 0.018 (9.10×; cached 0.020 vs 0.018, 1.08×; 0.20 vs 1.70); SG-Ter-Mer/Mat 0.167 vs 0.020 (8.54×; cached 0.023 vs 0.019, 1.24×; 0.18 vs 1.83); Full/Aug 0.412 vs 0.066 (6.45×; cached 0.072 vs 0.062, 1.15×; 0.21 vs 1.15); Full/Mat 0.424 vs 0.072 (6.08×; cached 0.084 vs 0.067, 1.32×; 0.23 vs 1.04). The selective median exceeds the role-blind median on all 612 instance–kernel records (paired Wilcoxon $p \leq 1.2 \times 10^{-10}$ per cell), and every operational family—34 in each class—has a positive median difference in both kernels. Metadata reconstruction accounts for 72–89% of the selective preprocessing time and at most 0.6 ms of the role-blind time. The batches contain no one-sided aborts, so the $10T_{\text{max}}$ abort sensitivity is identical to the primary convention. Each of the six dedicated CSVs contains both SA and Flat reruns from the same class–tolerance batch, independent of the main campaign. Adversarial signed-rank sensitivity for the four unrecorded pairs (Full augmented 0.1%, pre+solve endpoint): appending instances 045–048 with the adverse sign at above-maximal ranks gives $p = 0.47$ and $r_{\text{rb}} = +0.15$. For this sensitivity the nominal pool contains all 34 instances—the 27 observed nonzero pairs, the four appended adverse missing pairs and the three double timeouts restored as exact-zero operational ties—and the signed-rank procedure discards those three zeros, leaving an effective nonzero sample of $n = 31$, so the adjusted Full contrast is conditional on the 27 eligible observed pairs. This construction bounds the signed-rank evidence under adverse missing pairs; it is not a universal worst-case bound for the median ratio, the Hodges–Lehmann estimate, or the reliability-adjusted aggregate costs. On the same augmented sample the adversarial location estimates remain directionally favorable to selectivity (median SA/Flat 0.9596, HL 0.9465); the matrix-only counterpart moves the other way (median 1.0861, HL 1.0977, $p = 0.34$). The three double timeouts enter as operational ties (exact zeros).

Section S3. Traceability of the kernel-matched metadata grid

This section documents the experiment behind the kernel-matched metadata grid of the main text (the comparison of the matrix-only structure-aware rule SA-Mat against a same-kernel role-blind control), so the paper's central positive finding is fully reproducible.

Instance family. Every cell uses the block-heterogeneous coefficient family on $B = 4$ blocks, each with 6 continuous and 2 binary variables, 8 local rows and 3 coupling rows. Block i (1-indexed) carries a base magnitude 10^{e_i} with the exponents spread uniformly over $[-R, R]$, $e_i = -R + 2R\,(i-1)/(B-1)$, so that equal absolute coefficients denote different physical scales across blocks. The block-scale range R is the row variable of the grid and is set through the `SYNTH_BLOCK_RANGE` environment variable (default 2, which reproduces the original block-heterogeneous set byte-for-byte). At $R = 2$ the four block magnitudes are $10^{-2}, 10^{-2/3}, 10^{2/3}, 10^{2}$; at $R = 3$ they are $10^{-3}, 10^{-1}, 10^{1}, 10^{3}$. The coupling disturbance of severity S is applied on top of this, as in the main synthetic grid.

Seeds and variants. Each cell draws 30 instances from consecutive seeds starting at 20250803. The variants are Base, the matrix-only structure-aware rule SA-Mat, and its kernel-matched role-blind control Flat-Mat (the flat rule run with the coefficient-only kernel, added to the synthetic runner for this test); SA-Aug, Flat and Ruiz are included on the headline cell for reference. The exported constraint matrix of every variant is read back and decomposed with `scripts/spectral_audit.py`, and the reported statistic is the paired ratio $\kappa_2^+(SA - Mat)/\kappa_2^+(\text{Flat-Mat})$; because the two rules share both kernel and coverage, the ratio isolates the block metadata.

Headline cell. Table S13 gives the per-variant κ_2^+ distribution on the headline cell (coupling pattern, $R = 2$, $S = 6$, 30 seeds). The matrix-only structure-aware rule is lower than its role-blind twin on all 30 instances; the homogeneous control is the $[-1, 1]$ row of the metadata grid in the main text, where the two coincide within noise.

Table S13: Per-variant κ_2^+ on the headline cell of the kernel-matched grid (coupling pattern, block-scale range $[-2, 2]$, $S = 6$, 30 seeds): median, interquartile range, and full range. The paired SA-Mat/Flat-Mat ratio has median 0.098 (30/30 instances, Wilcoxon $p = 1.9 \times 10^{-9}$).

Variant	Median κ_2^+	IQR	Range
Base	5.3×10^8	$[2.4 \times 10^7,\ 3.2 \times 10^9]$	$[4.2 \times 10^5,\ 2.2 \times 10^{11}]$
SA-Mat	1.3×10^3	$[1.2 \times 10^3,\ 1.7 \times 10^3]$	$[8.7 \times 10^2,\ 1.7 \times 10^4]$
Flat-Mat	1.4×10^4	$[1.1 \times 10^4,\ 1.9 \times 10^4]$	$[6.2 \times 10^3,\ 2.1 \times 10^5]$
SA-Aug	2.1×10^5	$[4.3 \times 10^4,\ 6.1 \times 10^5]$	$[1.2 \times 10^4,\ 1.1 \times 10^7]$
Flat	2.4×10^4	$[1.8 \times 10^4,\ 3.1 \times 10^4]$	$[8.5 \times 10^3,\ 3.4 \times 10^5]$
Ruiz	3.9×10^5	$[8.3 \times 10^4,\ 9.2 \times 10^5]$	$[2.6 \times 10^4,\ 1.3 \times 10^7]$

Reproduction. The full 18-cell sweep and the per-instance κ_2^+ values are archived under `results-revision/spectral` (`metadata_grid.csv` for the sweep, `coupling_blockhet_spectral.csv` for the headline cell). One cell is produced by setting `SYNTH_BLOCK_RANGE=2` and piping to the solver the command `benchmarkSintetico --coef-family block_heterogeneous --volcar-lp <dir>` followed by the parameter lines `DummyLp`, seed `20250803`, `30` instances, `4` blocks, `6 2`, `8 3`, pattern `coupling`, severity `6`, variant list `Base,SA-Aug,SA-Mat,Flat,` and an output path; then `scripts/spectral_audit.py --lp-dir <dir>` reads the exported matrices. The whole grid is driven by `scripts/synth_metadata_grid.py`.

Section S4. Algorithmic summary

The recommended rule is the exhaustive matrix-only structure-aware scaling of Algorithm S1; the augmented variant studied for comparison differs only in folding the right-hand side into the magnitude extremes, as noted after it. Equation numbers refer to the main text.

Both assume an *exhaustive* role map—every row is either local to one block or a coupling row, $\mathcal{R}_{\text{all}} = \big(\dot{\bigcup}_{i \in \mathcal{B}} \mathcal{R}_i\big) \mathbin{\dot\cup} \mathcal{R}_{\text{cpl}}$, with no residual category. The archived operational campaigns used a non-exhaustive implementation in which a block of market-shortage rows remained unassigned and therefore unscaled; Algorithm S1 instead states the exhaustive rule that defines the target method, and the coverage-restored control of the main text removes the confound that non-exhaustive map introduced.

Augmented exploratory variant (SA-Aug). The augmented rule of the comparison differs from Algorithm S1 in two independent ways, not one. First, the per-row set $\mathcal{S}_r$ and the block extremes M_i, m_i fold in the nonzero right-hand side as well as the coefficients (their A, b counterparts, Eqs. (16), (20), (22) of the main text), and an optional local reliability floor $\varepsilon^{\text{loc}}_{\text{floor}}$ may gate the local update. Second, and independently, the per-row coupling factors $\gamma_r^A = 1/\widehat{\rho}_r^A$ are replaced by a single *collective* factor $\gamma^{A,b} = \arg\min_\gamma \Phi(\gamma)$ chosen by grid search over $\gamma \in [10^{-6}, 10^6]$ and applied to every coupling row whose proxy exceeds the threshold. Folding $|b_r|$ into the extremes is what carries the additional documented original-scale reliability failure mode on zero-right-hand-side balance rows; the collective coupling factor is the coarser treatment that obtains no spectral advantage over its role-blind control. We therefore report SA-Aug as exploratory and recommend the matrix-only rule above.

Section S5. Relocated results tables

These tables were moved here from the main text during the length revision; they carry detailed 0.1%-tolerance results, sensitivities, and audits that support but are not central to the argument. Each is referenced from the main text by its number here.

Algorithm S1 Exhaustive matrix-only structure-aware row scaling (recommended rule)

Require: Matrix A, right-hand side b, exhaustive row roles (local/coupling), block ownership $\beta(\cdot)$, coupling incidence
Ensure: Scaled matrix $\widetilde{A}$, scaled right-hand side $\widetilde{b}$

Initialize row factors $d_r \leftarrow 1$ for every row
for each local block $i \in \mathcal{B}$ **do**
  for each local row $r \in \mathcal{R}_i$ **do**
    Build $\mathcal{S}_r^A$ from the nonzero coefficients of row r (the right-hand side is excluded)
    if $\mathcal{S}_r^A \neq \emptyset$ **then**
      Compute $M_r^A = \max \mathcal{S}_r^A$ and $m_r^A = \min \mathcal{S}_r^A$
      Set $\alpha_r^A \leftarrow 1/\sqrt{M_r^A m_r^A}$
      if the safeguards admit α_r^A **then**
        Update $d_r \leftarrow d_r \alpha_r^A$
      end if
    end if
  end for
end for
Compute the locally scaled coefficients $\widetilde{a}_{rj} = d_r a_{rj}$
Compute block scales $s_i^A = \sqrt{M_i^A m_i^A}$ from the block coefficient extremes
for each coupling row $r \in \mathcal{R}_{\mathrm{cpl}}$ **do** ▹ per-row coupling factor (Section 4.3)
  Compute the coupling proxy $\widehat{\rho}_r^A = \left(\sum_j (|a_{rj}|/s_{\beta(j)}^A)^2\right)^{1/2}$
  if $\widehat{\rho}_r^A > 0$ **then**
    Set $\gamma_r^A \leftarrow 1/\widehat{\rho}_r^A$; clip γ_r^A to $[10^{-8}, 10^8]$
    if γ_r^A lies outside the near-identity band **and** passes the coefficient-window and reliability safeguards **then**
      Update $d_r \leftarrow d_r \gamma_r^A$
    end if
  end if
end for
Return $\widetilde{A} = \mathrm{diag}(d)A$ and $\widetilde{b} = \mathrm{diag}(d)b$

Table S14: At 0.1% MIPGap, completed-run Gurobi efficiency metrics retain the same descriptive pattern as at 1%; fixed-pool PAR10 is in the main-text reliability table.

Class	Variant	n	sgm T	med. sp.$^{\mathrm{det}}$	$\overline{\gamma}_{\mathrm{cb}}$
Simple	Base	195	8.28	1.00	**0.176**
	SA-Aug	195	5.65	**1.11**	0.180
	SA-Mat	198	**5.44**	1.10	0.182
	Ruiz	199	14.12	0.88	0.261
	Ruiz+Cols	195	12.55	1.02	0.199
Full	Base	31	305.30	1.00	0.027
	SA-Aug	30	**226.90**	1.26	0.028
	SA-Mat	31	306.39	1.12	0.025
	Ruiz	30	341.56	0.87	0.027
	Ruiz+Cols	31	267.10	**1.62**	**0.022**
SG-Ter-Mer	Base	62	14.49	1.00	0.642
	SA-Aug	63	8.99	1.32	0.473
	SA-Mat	65	**7.97**	**1.35**	**0.439**
	Ruiz	65	14.27	1.05	0.633
	Ruiz+Cols	62	19.16	0.78	0.625

Notes. Bold values mark the best descriptive value within each class and metric; lower is better for sgm T and $\overline{\gamma}_{\mathrm{cb}}$, higher is better for med. sp.$^{\mathrm{det}}$.

Table S15: At 0.1% MIPGap, CPLEX remains close to parity for the structure-aware variants, reinforcing the contrast with the Gurobi results in the tested pipelines.

Class	Variant	n	sgm T	med. sp.$^{\mathrm{det}}$	$\overline{\gamma}_{\mathrm{cb}}$
Simple	Base	204	5.14	1.00	1.2×10^{-12}
	SA-Aug	204	4.20	1.02	6.5×10^{-13}
	SA-Mat	204	4.48	1.01	5.6×10^{-13}
	Ruiz	204	**4.13**	1.00	$\mathbf{1.0 \times 10^{-15}}$
	Ruiz+Cols	204	5.28	0.94	7.7×10^{-13}
Full	Base	33	421.59	1.00	2.7×10^{-8}
	SA-Aug	34	422.62	1.07	2.7×10^{-8}
	SA-Mat	33	370.45	1.05	2.3×10^{-8}
	Ruiz	34	**285.10**	**1.28**	$\mathbf{1.9 \times 10^{-8}}$
	Ruiz+Cols	33	377.41	0.96	5.2×10^{-8}
SG-Ter-Mer	Base	68	2.82	1.00	1.1×10^{-10}
	SA-Aug	68	2.84	1.01	$\mathbf{1.0 \times 10^{-10}}$
	SA-Mat	68	2.84	1.00	1.1×10^{-10}
	Ruiz	68	**2.61**	**1.02**	1.1×10^{-10}
	Ruiz+Cols	68	4.02	0.85	1.3×10^{-10}

Notes. Bold values mark the best descriptive value within each class and metric. Under CPLEX these descriptive minima do not imply a robust positive external-scaling effect.

Table S16: Descriptive postselected Gurobi internal-scaling comparison at 1% MIPGap. Within each class, the internal mode with the lowest class-level solver-time sgm is selected on the same data; SA-Aug and SA-Mat use default internal scaling. Ratios are medians of $T_{\text{internal}}/T_{\text{SA}}$ under the main paired convention: double non-completions are excluded and a one-sided non-completion is assigned $T_{\max} = 3600\,\text{s}$. This same-sample postselection is descriptive, not inferential.

Class	Selected mode	sgm T_{int}	sgm T_{Aug}	sgm T_{Mat}	n_{Aug}	n_{Mat}	Med. ratio Aug	Med. ratio Mat
Simple	aggressive	3.79	2.13	2.17	204	204	0.97	0.96
SG-Ter-Mer	equilibration	14.66	8.28	8.64	67	67	1.18	1.28
Full	equilibration	128.45	95.24	102.03	32	32	1.21	1.25